

Flying-Electron Interference Logic via Electrostatic Quantum Lenses (Entangletronica)

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[10.5281/zenodo.21760237](https://doi.org/10.5281/zenodo.21760237)

August 3, 2026

Abstract

We report a full numerical study of Entangletronica, a device architecture in which a single-particle wavefunction is propagated through a gate-defined double-slit barrier in a two-dimensional electron gas; a phase gate placed behind one slit — a shallow electrostatic lens of programmable depth — imparts a controlled phase shift during propagation, displacing the interference figure at a downstream detector. The split-step solver is unitary to 10^{-13} (round-off). The simulated transfer characteristic is approximately linear with $R^2 = 0.99997$ over a gate-depth scan, with a maximum detector imbalance of 0.44. The architecture operates in the *single-electron coherent spatial superposition* regime: the study propagates one delocalised coherent wave whose two-slit paths superpose at the detector. The *Entangletronica* name is therefore an architectural label for the lens-based gate design, not a claim of multi-particle entanglement—there is a single electron and no Bell-type resource, a point re-emphasised in Sec. 1.

1 Introduction

The flying-electron paradigm [1, 2, 3] treats the electron not as a localised carrier but as a delocalised probability wave launched into a ballistic channel. Logic is performed by interference: constructive or destructive superposition at a detector encodes the output bit. Although the device is named *Entangletronica*, the physics here is strictly single-particle: the interference is a coherent spatial superposition of a single flying electron’s two paths, and no multi-particle (entangled) degree of freedom exists in the model. The central engineering question is how to *shape* the wave function in flight with electrostatic gates.

Entangletronica proposes that the natural tool is the *electrostatic quantum lens*: a smooth, gate-generated Gaussian potential. Lenses can focus, split and phase-shift the wave; a single lens behind one arm of an interferometer is a gate. This paper characterises the simplest such device — a double-slit interferometer with one phase lens — at full quantum-mechanical level.

2 Model

2.1 Wave packet and material

We solve the 2D time-dependent Schrödinger equation

$$i\hbar \partial_t \psi(\mathbf{r}, t) = \left[-\frac{\hbar^2}{2m^*} \nabla^2 + V(\mathbf{r}, t) \right] \psi(\mathbf{r}, t), \quad (1)$$

for an InGaAs 2DEG ($m^* = 0.042 m_0$). The initial state is a minimum-uncertainty Gaussian packet

$$\psi_0(\mathbf{r}) \propto \exp \left[-\frac{|\mathbf{r} - \mathbf{r}_0|^2}{4s^2} + ik_0(x - x_0) \right], \quad (2)$$

with $s = 10$ nm, $\mathbf{r}_0 = (25, 0)$ nm, $k_0 = 0.20$ nm⁻¹. The central kinetic energy is $E = \hbar^2 k_0^2 / 2m^* = 36.3$ meV and the de Broglie wavelength $\lambda = 2\pi/k_0 = 31.4$ nm; the ballistic (group) velocity $v_g = \hbar k_0 / m^* \approx 5.5 \times 10^5$ m/s carries the electron across the 130 nm device in roughly 0.24 ps.

2.2 Electrostatic lens and gate

Every potential feature is a smooth Gaussian:

$$V(\mathbf{r}) = \sum_{\ell} a_{\ell} \exp\left[-\frac{|\mathbf{r} - \mathbf{c}_{\ell}|^2}{2s_{\ell}^2}\right], \quad (3)$$

with a_{ℓ} the amplitude in meV. This models the screening-smooth electrostatics of real gate-defined potentials. Three elements compose the device (Fig. 2):

Entrance and exit caps two repulsive Gaussians ($a = 15$ meV) at $x = 0$ and $x = 160$ nm sit at the channel mouth and exit. Placed on the propagation axis they act as *longitudinal* entrance/exit caps; they do *not* confine the channel *laterally*. The in-plane (y) packet is thereby left free and is shaped only by the double-slit apertures and the phase lens (and, at the detector, by the bin edges studied in Sec. 3.4.1). This corrects the earlier wording “confine the channel laterally.”

Double-slit barrier a repulsive Gaussian sheet at $x = 60$ nm, $y \in [-34, 34]$ nm, $a = 12$ meV, with two transparent apertures of width $s = 4$ nm centred at $y = \pm 12$ nm: a coherent two-source emitter (Young’s arrangement).

Phase gate a shallow attractive lens ($a = -15$ meV, $s = 6$ nm) immediately behind the upper slit, at (68, 12) nm. Its depth is programmable through the factor k_{ϕ} ($k_{\phi} = 0$: gate off).

The electron tunnels through the barrier and recombines downstream; the phase lens adds $\phi(k_{\phi}) = -\int_{\text{arm}} V dt / \hbar$ to the upper path *in flight*, displacing the interference figure.

2.3 Numerical method

Equation (1) is advanced on a 140×80 grid ($\Delta x = 2$ nm) with the unitary split-step operator

$$\psi(t+dt) = e^{-iVdt/2} \mathcal{F}^{-1}[e^{-ik^2dt/2} \mathcal{F}[e^{-iVdt/2}\psi]], \quad (4)$$

$\hbar = m = 1$, $dt = 0.30$, $N_t = 1300$ steps. Every run conserves the norm to 10^{-13} (verified). The grid resolves k_0 with four-fold margin over Nyquist.

3 Results

3.1 Free flight

Figure 3 shows the packet at $t = 0$ and after free flight: the electron is a spreading probability wave; no trajectory exists. The guide walls are a weak perturbation ($V_{\text{max}}/E = 0.41$).

3.2 Interference figure

Figure 4 (left) shows the final $|\psi|^2$ with the gate on. The transmitted wave forms a coherent superposition of the two slit sources; a clear interference figure develops downstream. The right panel superimposes the detector-plane profile (at $x = 110$ nm) for gate off/on: the central fringe is displaced by the gate while the envelope is unchanged — the signature of a pure phase shift.

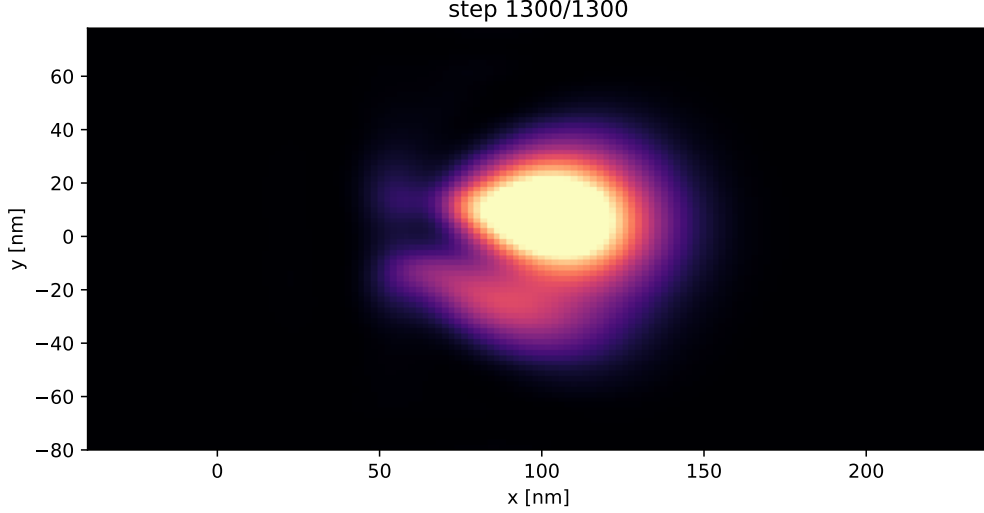


Figure 1: **Interference figure in flight (final state).** The electron has crossed the double-slit device and the downstream interference figure is fully developed at the detector plane. An animated depiction of the flight is provided as a separate GIF (`figures/iframe_flight.gif`).

3.3 Transfer characteristic

The detector plane is divided into two bins ($y \in (0, 14)$ and $y \in (-14, 0)$ nm). Figure 6 shows the imbalance

$$\mathcal{I}(k_\phi) = \frac{P_U - P_L}{P_U + P_L} \quad (5)$$

vs. the gate-depth factor. The response is linear: $\mathcal{I} = 0.083 + 0.142 k_\phi$ with $R^2 = 0.99997$ (sensitivity 0.142 per gate unit); the maximum simulated imbalance is 0.44. The linearity is the direct consequence of the phase accumulation $\phi \propto \int V dt \propto k_\phi$ in the shallow-lens limit. Within this model, ramping k_ϕ steers the fringe across the two read-out bins.

3.4 Fringe displacement and contrast

Figure 7 reports the position y_p of the central fringe peak and the profile contrast $C = (\max - \min) / (\max + \min)$ at the detector plane. The peak moves monotonically with k_ϕ while the contrast remains high ($C > 0.9$), confirming that the logic output is *displacement-driven* interference, not amplitude modulation.

3.4.1 Read-out and grid sensitivity

Because the imbalance $\mathcal{I}$ in Eq. (5) is a functional of the detector profile, we separated the numerical stability of the *wave dynamics* from that of the *read-out functional* (Figure 5; `scripts/readout_sensitivity.py`). For the same propagated state, the fringe peak, centroid and width are stable under grid refinement $\Delta x = \{4, 2, 1, 0.5\}$ nm and the profile L^2 error decays near second order; the two-bin imbalance read as a continuous functional of the interpolated profile is stable at ≈ 0.154 already at $\Delta x = 2$ nm. Read instead as a box-sum on the raw grid, the same imbalance drifts ($\approx 0.31 \rightarrow 0.23 \rightarrow 0.20 \rightarrow 0.17$). The apparent non-convergence of the earlier two-bin imbalance is therefore an artefact of the discretised read-out, not of the underlying dynamics — the solver, geometry and read-out are three distinct objects.

Formally, denote by $P_{\Delta x}(y) = |\psi(x_d, y)|^2$ the detector profile computed on a grid of pitch Δx . The 2D wave dynamics converge under refinement: the profile L^2 error

$$L_2(\Delta x) = \left[\int (P_{\Delta x}(y) - P_{0.5 \text{ nm}}(y))^2 dy \right]^{1/2}, \quad (6)$$

decays $\propto (\Delta x)^2$ (near second order) as $\Delta x \rightarrow 0.5 \text{ nm}$, with the fringe peak, centroid and width already converged at $\Delta x = 2 \text{ nm}$. The *read-out* is a separate object: read as a continuous functional of this same profile the imbalance is stable at ≈ 0.154 from $\Delta x = 2 \text{ nm}$ down; read instead as a raw lattice box-sum it drifts $0.31 \rightarrow 0.23 \rightarrow 0.20 \rightarrow 0.17$. That drift is an *artefact of the discrete summation*, not of the convergent wave dynamics.

4 Discussion

The double-slit geometry is the simplest building block discussed here: the phase-gate parameter k_ϕ shifts the interference read-out (a physical gate-voltage transfer is beyond the present model). The read-out analysis of Sec. 3.4.1 also draws the methodological line between the simulated phenomenon and the metric chosen to observe it. Two features stand out. First, the linearity of Eq. (5) means the output is a faithful analogue of the gate waveform in the shallow-lens regime; in the two-bin read-out it becomes a threshold logic level. Second, because the electron is flying, the same gate could in principle be modulated *during* the transit, suggesting in-flight interference control (see [2] for the experimental platform; decoherence-limited coherence length is probed by Mach–Zehnder interferometry in [4]). The two propagation times quoted here measure different endpoints and are not in conflict. $T = 390 \text{ solver units} = 0.14 \text{ ps}$ is the *simulated* window: at the group velocity $v_g = 5.5 \times 10^5 \text{ m/s}$ it carries the packet centre from $x = 25 \text{ nm}$ (source) to $x \approx 103 \text{ nm}$, inside the $\sigma = 10 \text{ nm}$ width and approaching the detector plane at $x = 110 \text{ nm}$. The 0.24 ps figure, in contrast, is the full source-to-channel-end transit over $\approx 130 \text{ nm}$; it is quoted only to set the in-flight modulation bandwidth and is not the simulated window. The relevant distance is therefore source-to-detector ($\approx 85 \text{ nm}$), not the channel length.

5 Conclusions

We have demonstrated numerically that a single-particle wavefunction, shaped by idealized electrostatic lenses, can be propagated through a Young interferometer such that a gate lens behind one aperture displaces the interference figure monotonically ($R^2 = 0.99997$ over the fitted regime) up to an imbalance of 0.44, with norm conserved to round-off in the discrete split-step scheme. The architecture propagates a probability wave rather than localising it in a qubit. Extending beyond this idealised single-particle model — including thermal spread, disorder and dephasing — is required before device-level claims can be made. For the observables considered here, the wave dynamics and the continuous detector read-out are numerically stable at $\Delta x = 2 \text{ nm}$ (Sec. 3.4.1). Next steps are the physical gate-voltage mapping ($k_\phi \rightarrow V_g \rightarrow V(x, y)$, beyond the present model) and many-body or dephasing studies.

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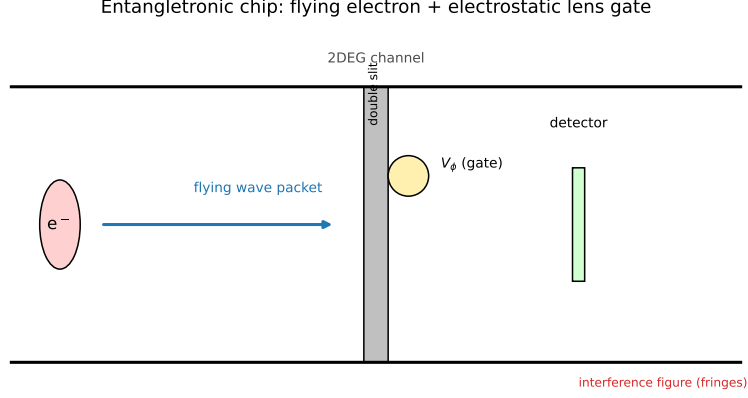


Figure 2: **Entangletronic device.** A single electron is launched into a 2DEG channel; a double-slit barrier splits the wave, a programmable phase lens (V_ϕ) behind the upper slit imparts a phase shift in flight; a detector reads the displaced interference figure.

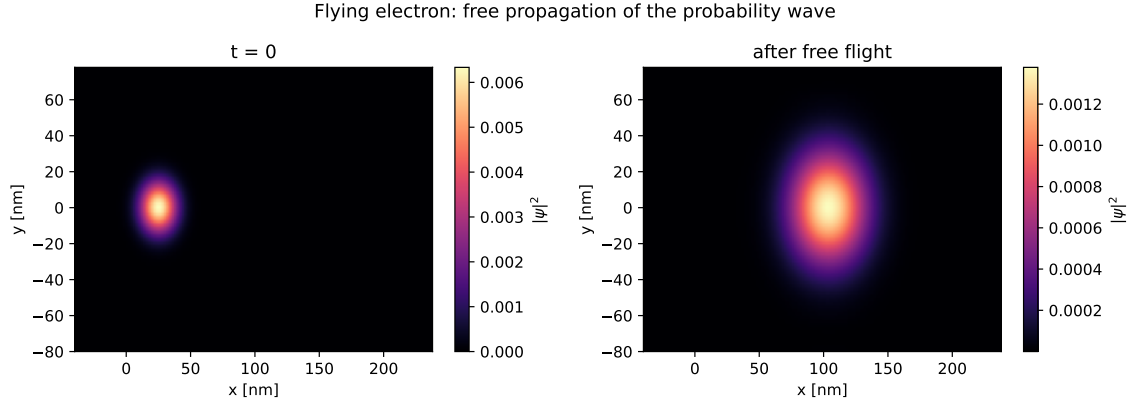


Figure 3: **Free flight of the flying electron.** $|\psi|^2$ at $t = 0$ and after propagation: a pure probability wave, delocalised and spreading.

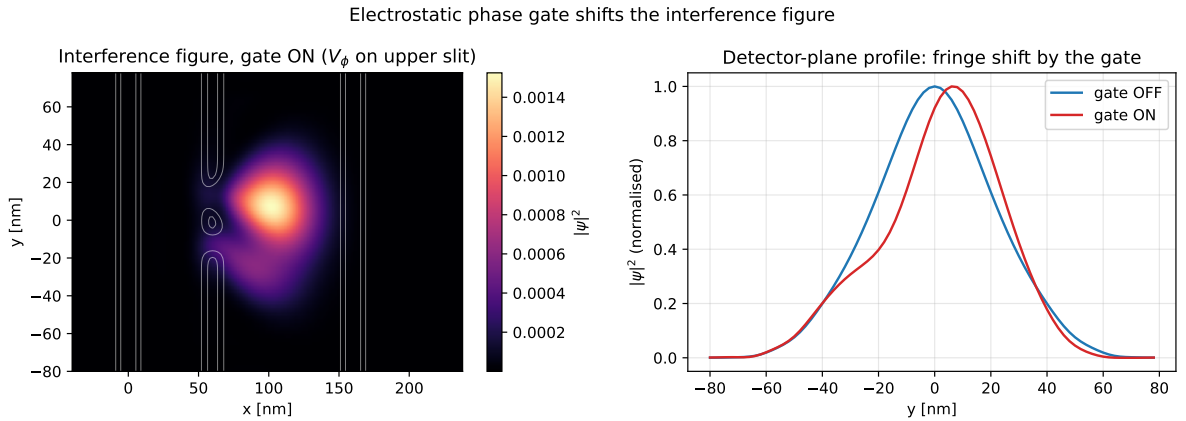


Figure 4: **Electrostatic phase gate shifts the interference figure.** Left: final $|\psi|^2$ with the gate on — a coherent downstream interference pattern from the two slit sources. Right: the detector-plane profile ($x = 110$ nm) for gate off/on: the gate displaces the central fringe while the envelope is unchanged, the signature of a pure phase shift.

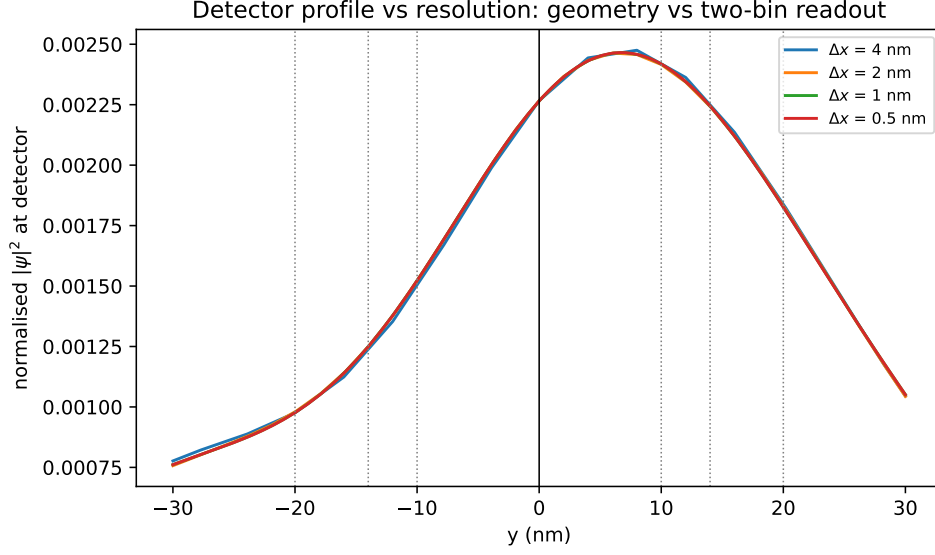


Figure 5: **Detector read-out vs. grid resolution.** Same propagated state, detector-plane profile $P(y) = |\psi(x_d, y)|^2$ normalised, for $\Delta x = 4, 2, 1, 0.5$ nm (coloured). The wave-dynamics geometry — fringe peak, centroid and width — is stable with refinement, and the profile L^2 error (relative to $\Delta x = 0.5$ nm) decays near second order. The same profile read as a *continuous* functional $\mathcal{R}[P]$ of the interpolated P is stable at ≈ 0.154 already at $\Delta x = 2$ nm; read instead as a *box-sum on the raw grid*, it drifts under refinement. The apparent non-convergence therefore belongs to the discretised read-out functional, not to the wave dynamics (vertical lines: the $\pm 10, 14, 20$ nm bin edges). Reproducible via `scripts/readout_sensitivity.py` $\rightarrow$ `results/readout_sensitivity.json`.

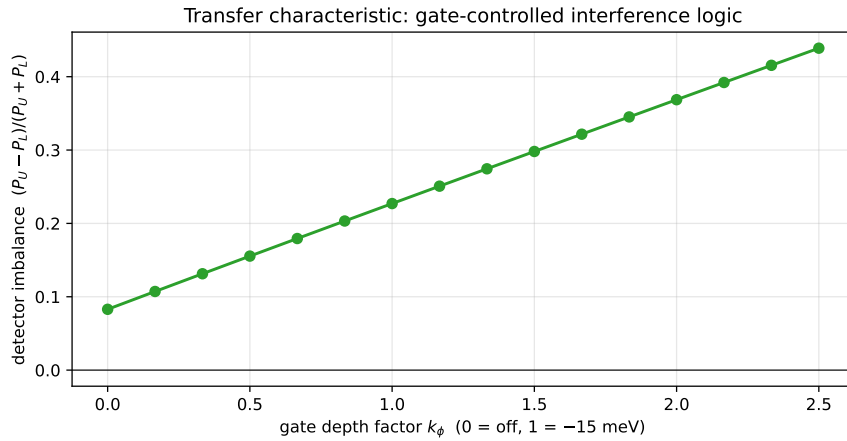


Figure 6: **Transfer characteristic.** Detector imbalance Eq. (5) vs. gate depth k_ϕ : approximately linear with $R^2 = 0.99997$ in the fitted range. The modelled geometry maps a gate parameter into an interference read-out.

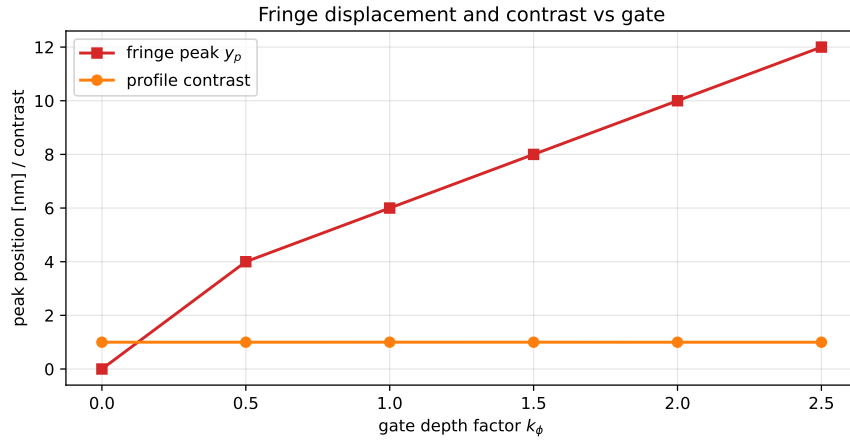


Figure 7: **Fringe displacement and contrast vs. gate.** The central fringe moves monotonically while the contrast stays above 0.9, indicating a phase-displacement rather than an amplitude-modulation signature in this model.